3.2 Linear Programming

A linear cost, linear inequalities, and a dual that is also an LP.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

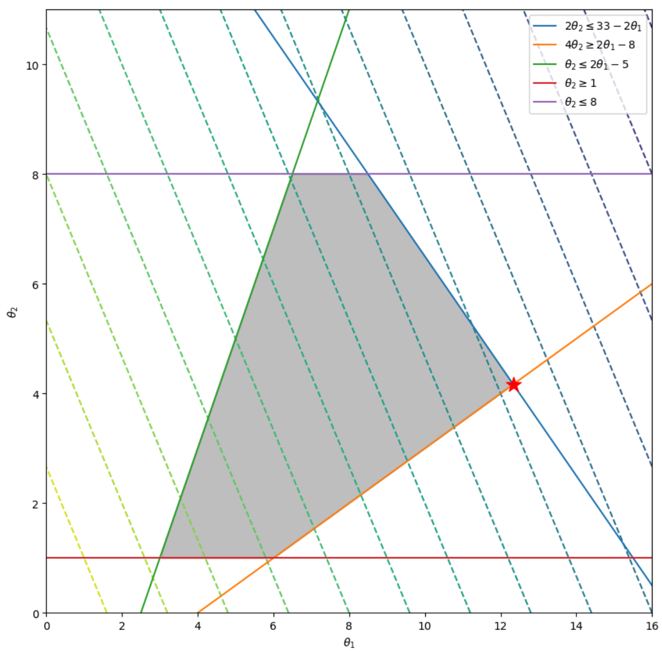
The figures follow Deisenroth, Faisal, and Ong, *Mathematics for Machine Learning*.

1. Linear cost, linear inequalities

A **linear program** is the special case where J and the constraints are linear. In machine learning that is how you optimize a linear model under linear limits, and how you write some feature-selection and dimension-reduction tasks so a solver can hit the feasible set.

Level sets of a linear cost are parallel hyperplanes. Inequality constraints are half-spaces. The feasible set is a polyhedron (shaded). The min, if it exists, sits at a vertex or on a face.

$$\begin{aligned} \min_{\boldsymbol{\theta}} \quad & -5\theta_1 - 3\theta_2 \\ \text{subject to} \quad & 2\theta_1 + 2\theta_2 \le 33 \\ & 2\theta_1 - 4\theta_2 \le 8 \\ & -2\theta_1 + \theta_2 \le -5 \\ & -\theta_2 \le -1 \\ & \theta_2 \le 8 \end{aligned}$$



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2. The primal

$$\min_{\theta \in \mathbb{R}^d} \mathbf{c}^\top \theta \quad \text{subject to} \quad A\theta \le \mathbf{b},$$

with $A \in \mathbb{R}^{m \times d}$ and $\mathbf{b} \in \mathbb{R}^m$. That is d variables and m linear constraints.

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3. Dual of an LP

Lagrangian, with $\lambda \geq 0$,

$$\mathcal{L}(\theta, \lambda) = \mathbf{c}^{\top}\theta + \lambda^{\top}(A\theta - \mathbf{b}) .$$

Collect terms in θ:

$$\mathcal{L}(\theta, \lambda) = \left(\mathbf{c} + A^{\top}\lambda\right)^{\top}\theta - \lambda^{\top}\mathbf{b} .$$

Stationarity in θ forces

$$\mathbf{c} + A^{\top}\lambda = 0 .$$

Then $D(\lambda) = -\lambda^{\top}\mathbf{b}$, and the dual is itself an LP with m variables:

$$\max_{\lambda \in \mathbb{R}^m} -\mathbf{b}^{\top}\lambda \quad \text{subject to} \quad \mathbf{c} + A^{\top}\lambda = 0, \quad \lambda \geq 0 .$$

Solve the **primal** if d is small. Solve the **dual** if m is small. Same optimal value when the LP is feasible and bounded.

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4. Practice

1. In the figure, why is the minimizer of a feasible bounded LP at a vertex (or on a face), not in the interior of the polyhedron?
2. After forming the Lagrangian, what equation knocks θ out and leaves a dual only in λ?
3. You have few variables and many constraints. Do you prefer the primal or the dual, and why?